

FB_Inference - SCL Excerpt (B-spline LUT)

```
FUNCTION_BLOCK FB_Inference

VAR_INPUT features:ARRAY[0..27] OF REAL; END_VAR

VAR_OUTPUT class_id:INT; confidence:REAL; END_VAR

FOR i:=0 TO 27 DO

    lo:=0;

    FOR j:=1 TO 13 DO

        IF features[i]>=W_DB.g0[j] THEN lo:=j; END_IF;

    END_FOR;

    t_val:=(features[i]-W_DB.g0[lo])/(W_DB.g0[lo+1]-W_DB.g0[lo]+1E-10);

    FOR o:=0 TO 15 DO

        v3[o*28+i]:=W_DB.t1[base+lo]*(1-t_val)+W_DB.t1[base+lo+1]*t_val;

    END_FOR;

END_FOR;

END_FUNCTION_BLOCK
```

Syntax-highlighted SCL, rendered as print-ready vector figure.